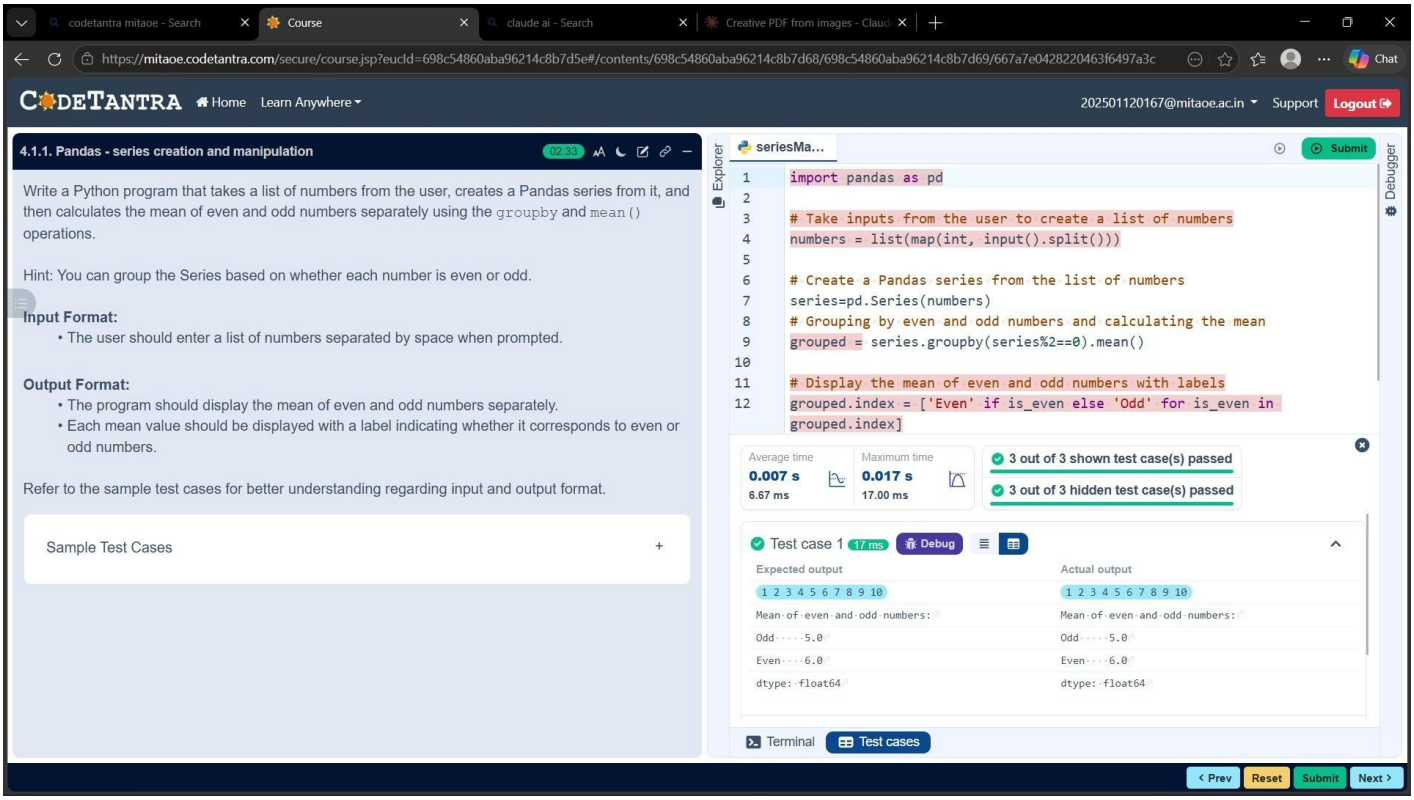
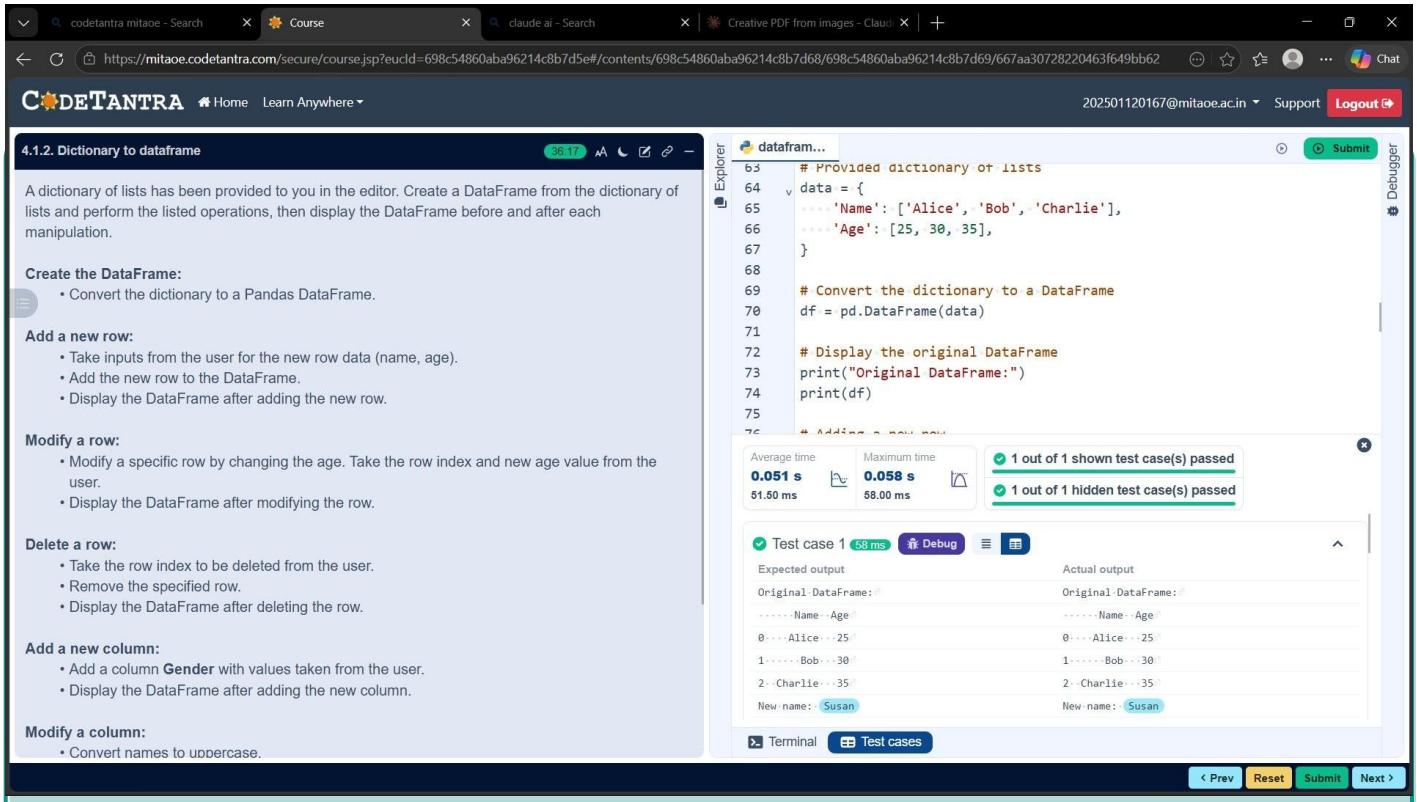
Creating a Pandas Series from user input and calculating mean of even/odd numbers using groupby and mean().



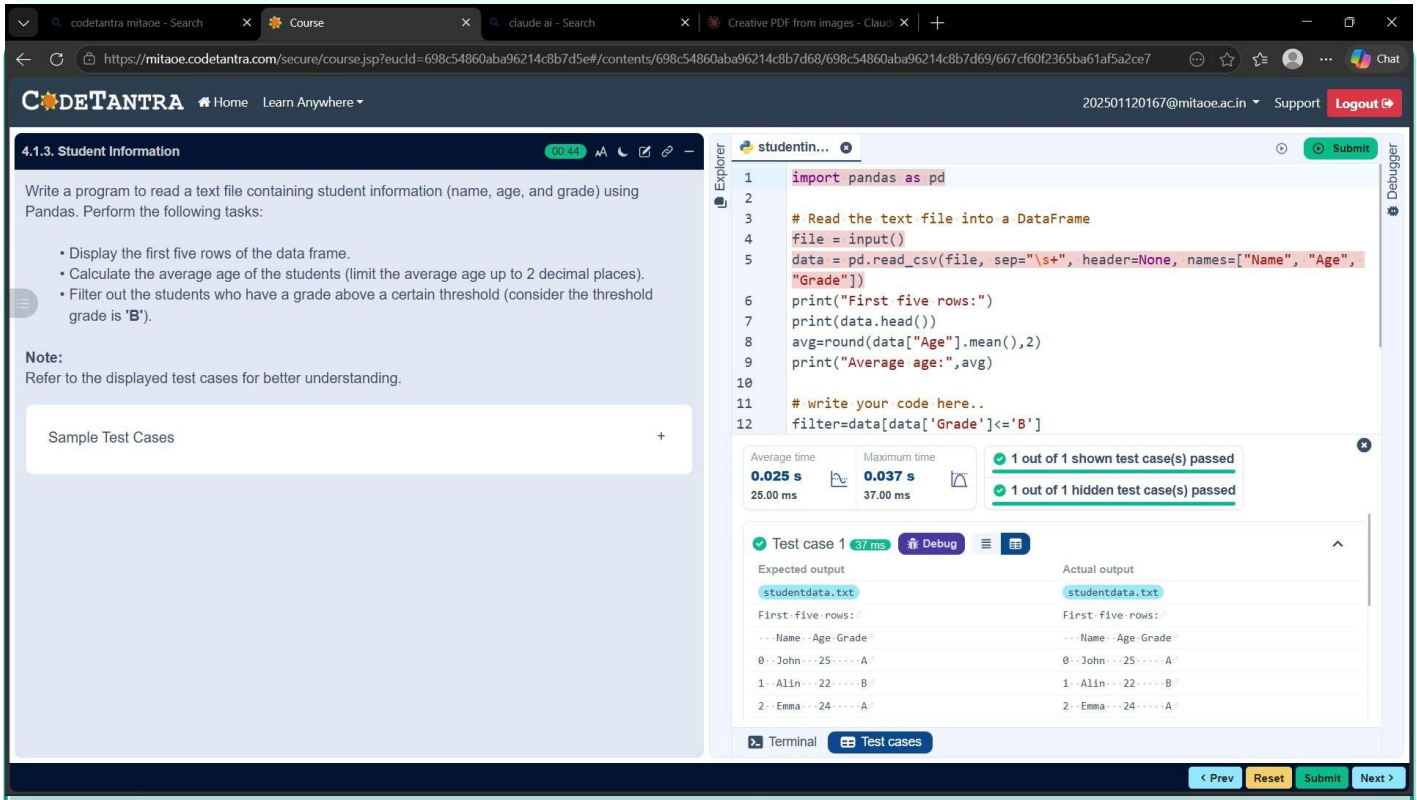
Converting a dictionary of lists into a DataFrame; adding/modifying/deleting rows and columns.



Exercise 4.1.3 · Student Information

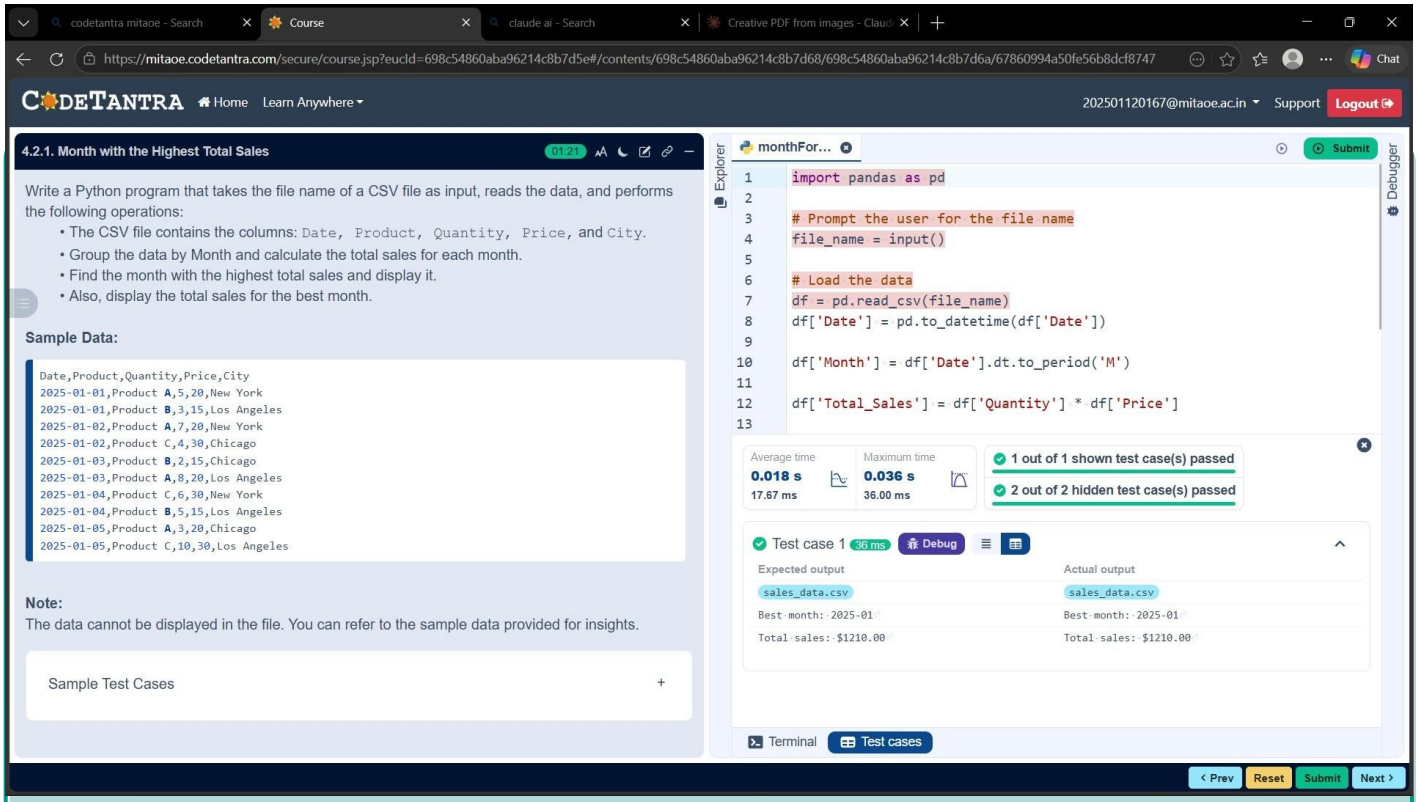
Reading a text file into a DataFrame, displaying first 5 rows, calculating average age, and filtering by grade.

✓ All Tests Passed



Exercise 4.2.1 · Month with the Highest Total Sales

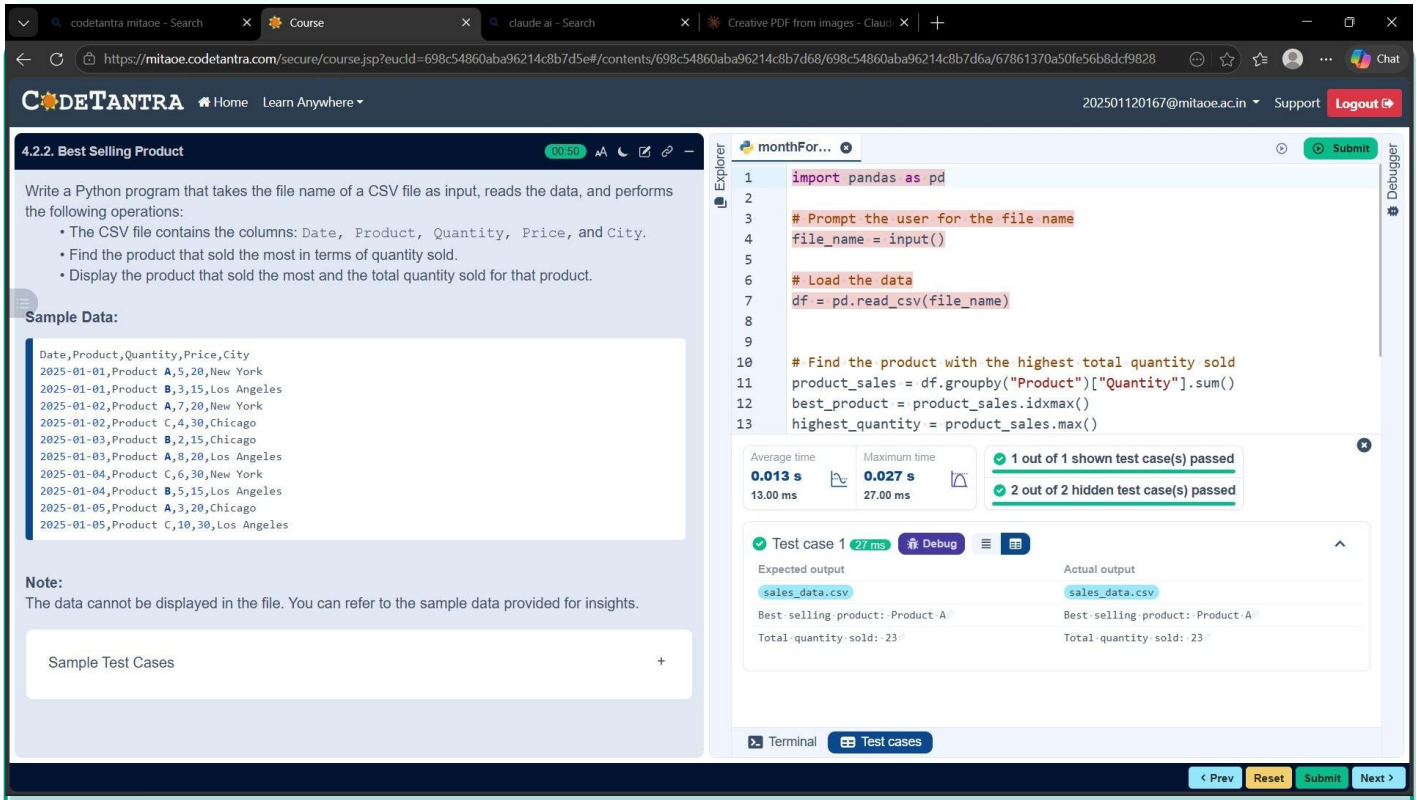
Grouping CSV sales data by month, computing total sales per month, and identifying the best-performing month.



Exercise 4.2.2 · Best Selling Product

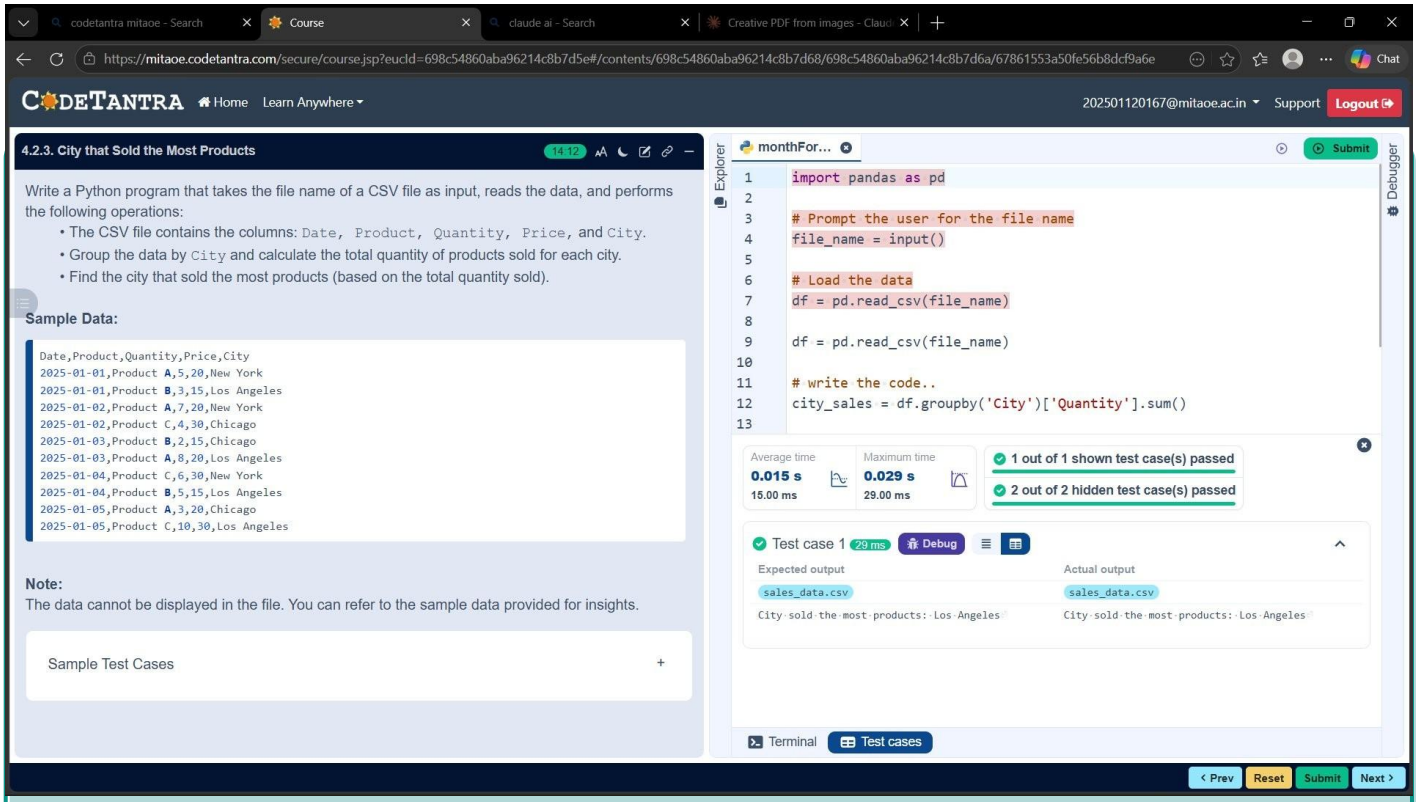
Finding the product with the highest total quantity sold across all transactions in the dataset.

✓ All Tests Passed



Exercise 4.2.3 · City that Sold the Most Products

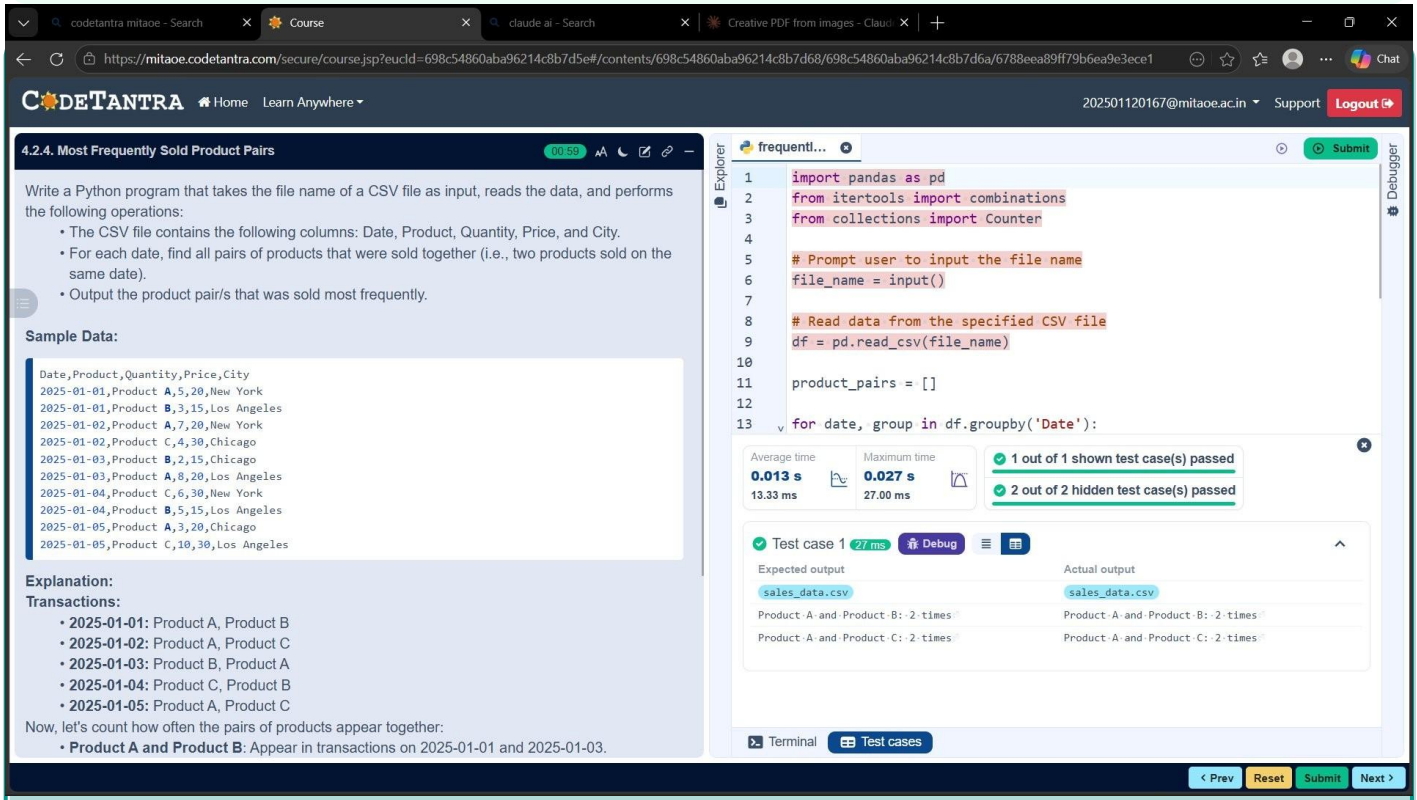
Grouping sales data by City and finding which city sold the greatest total quantity of products.



Exercise 4.2.4 · Most Frequently Sold Product Pairs

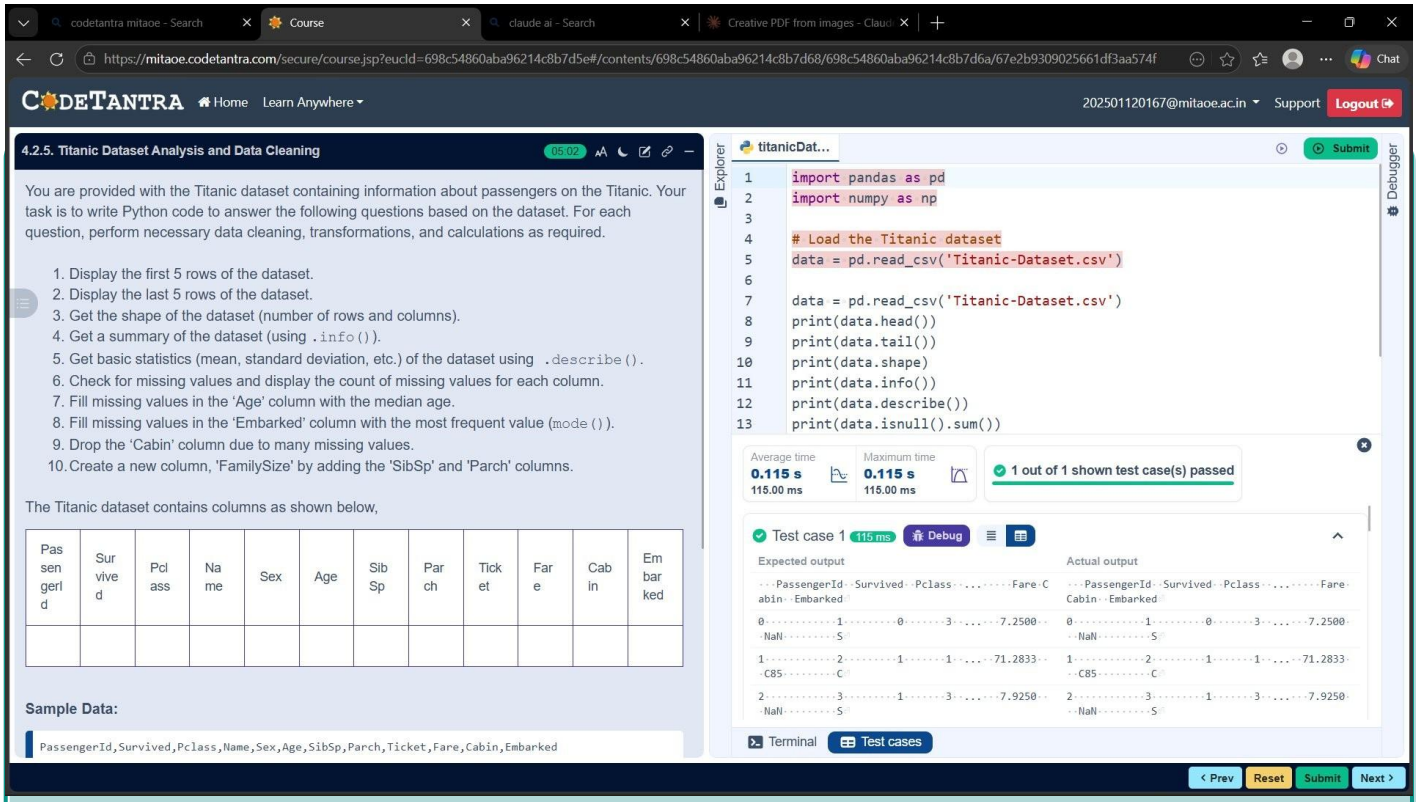
Using itertools and Counter to find product pairs sold together most often on the same date.

✓ All Tests Passed



Exercise 4.2.5 · Titanic Dataset Analysis & Cleaning

Loading the Titanic CSV, displaying head/tail, checking shape, info, describe, missing values, and feature engineering.



Exercise 4.2.7 · Titanic Dataset Analysis - Part 3

Survival rates by class, embarkation, family size, and IsAlone; average fare and age by class and survival status.

✓ All Tests Passed

CODETANTRA Home Learn Anywhere 202501120167@mitaoe.ac.in Support Logout

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3 03:23

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Calculate the survival rate by class.
2. Calculate the survival rate by embarkation location (Embarked_S).
3. Calculate the survival rate by family size (FamilySize).
4. Calculate the survival rate by being alone (IsAlone).
5. Get the average fare by passenger class (Pclass).
6. Get the average age by passenger class (Pclass).
7. Get the average age by survival status (Survived).
8. Get the average fare by survival status (Survived).
9. Get the number of survivors by class (Pclass).
10. Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC 17599,71.2833,C85,C
```

titanicDat... Submit

```python
import pandas as pd
import numpy as np

# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)

print(data.groupby('Pclass') ['Survived'].mean())
#2. Calculate the survival rate by embarked location (Embarked_S)
print(data.groupby('Embarked_S') ['Survived'].mean())
#3. Calculate the survival rate by family size
```

Average time 0.056 s 56.00 ms | Maximum time 0.056 s 56.00 ms | 1 out of 1 shown test case(s) passed

Test case 1 56 ms Debug

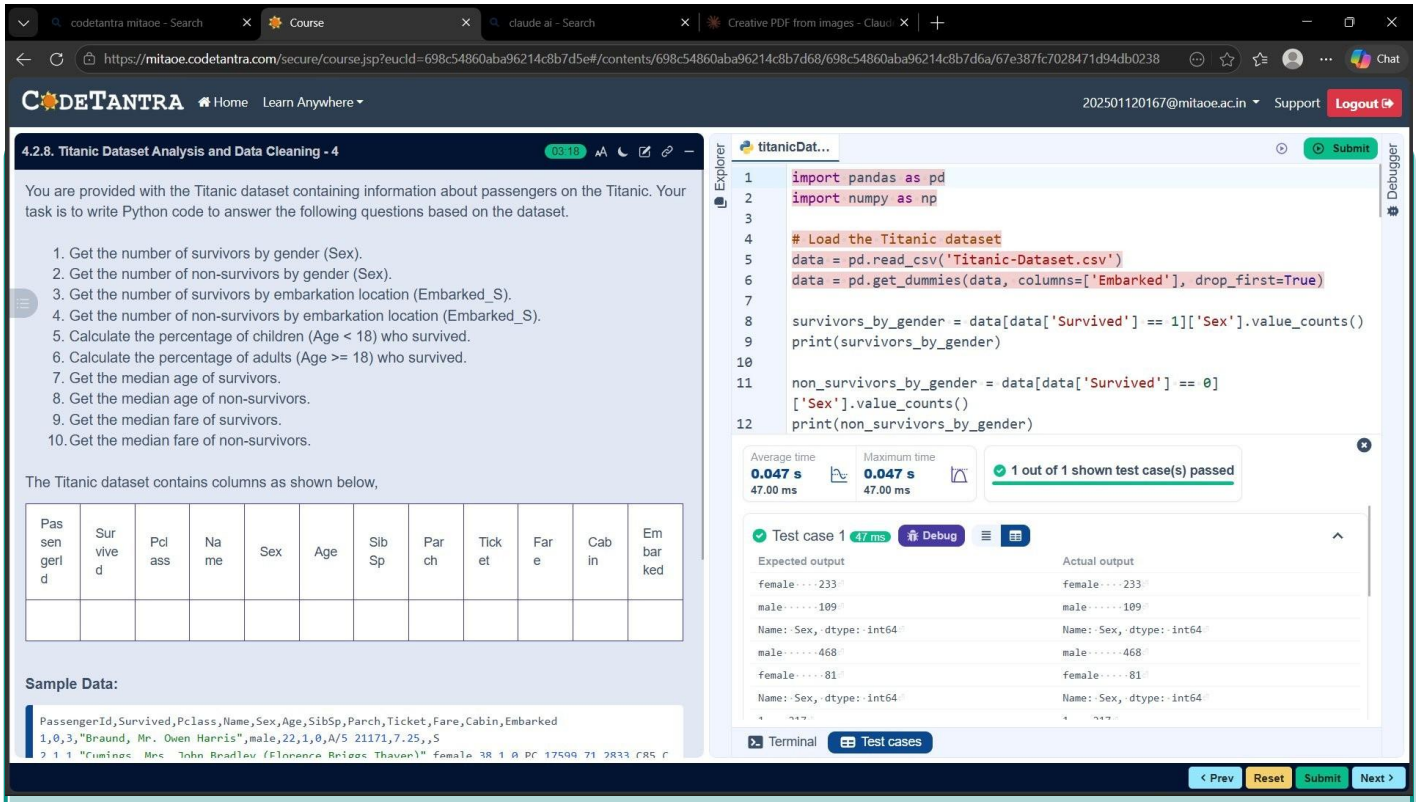
Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S

Terminal Test cases

Prev Reset Submit Next

Exercise 4.2.8 · Titanic Dataset Analysis - Part 4

Survivors/non-survivors by gender and embarkation; child vs adult survival %; median age and fare by survival.


codetantra mitaoe - Search
Course
claude ai - Search
Creative PDF from images - Claud
https://mitaoe.codetantra.com/secure/course.jsp?eucId=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d68/698c54860aba96214c8b7d6a/67e387fc7028471d94db0238
Chat
CODETANTRA
Home
Learn Anywhere
202501120167@mitaoe.ac.in
Support
Logout
4.2.8. Titanic Dataset Analysis and Data Cleaning - 4
03:18
You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.
1. Get the number of survivors by gender (Sex).
2. Get the number of non-survivors by gender (Sex).
3. Get the number of survivors by embarkation location (Embarked_S).
4. Get the number of non-survivors by embarkation location (Embarked_S).
5. Calculate the percentage of children (Age < 18) who survived.
6. Calculate the percentage of adults (Age >= 18) who survived.
7. Get the median age of survivors.
8. Get the median age of non-survivors.
9. Get the median fare of survivors.
10. Get the median fare of non-survivors.
The Titanic dataset contains columns as shown below,
Passengerld
Survived
Pclass
Name
Sex
Age
SibSp
Parch
Ticket
Fare
Cabin
Embarked
Sample Data:
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
Explorer
titanicDat...
Submit
Debugger
import pandas as pd
import numpy as np
# Load the Titanic dataset
data = pd.read_csv('Titanic-Dataset.csv')
data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
survivors_by_gender = data[data['Survived'] == 1]['Sex'].value_counts()
print(survivors_by_gender)
non_survivors_by_gender = data[data['Survived'] == 0]['Sex'].value_counts()
print(non_survivors_by_gender)
Average time
0.047 s
47.00 ms
Maximum time
0.047 s
47.00 ms
1 out of 1 shown test case(s) passed
Test case 1
47 ms
Debug
Expected output
Actual output
female 233
male 109
Name: Sex, dtype: int64
male 468
female 81
Name: Sex, dtype: int64
Terminal
Test cases
Prev
Reset
Submit
Next